Unified Computational Benchmark for First-Passage Problems Luca / Generating-Function Family vs. Time-Domain Recursion Family

valley-k-small automated report

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Abstract

This document is the single active computational-method comparison report in the repository. It keeps the scientific reports separate, but collapses the comparison line into one bilingual benchmark with six workloads. The transform-domain side is treated explicitly as a *family* rooted in the propagator, renewal, and bounded-heterogeneity formalisms developed by Giuggioli, Sarvaharman, and collaborators, rather than as one monolithic single-paper method [1, 2, 3, 4]. The benchmark rule is *practical native-task fairness*: each report is timed on the task it actually needed to complete, rather than on an artificially synchronized full-tail horizon. Under the diagnostic protocol the median ratio $t_{\text{time}}/t_{\text{Luca}}$ is 0.0490; time recursion wins on ENC-FIXED, ENC-ANY, TT-C1, REF-S0, while Luca/GF wins on RING-1T-paper, TT-LF1. Under the curve protocol the median ratio is 0.0398; time recursion wins on ENC-FIXED, ENC-ANY, TT-C1, REF-S0, while Luca/GF wins on RING-1T-paper, TT-LF1.

1 Goal and Comparison Rule

We compare only two families.

- **Luca / generating-function family**: closed-form or defect-free propagators, finite-support heterogeneity corrections, renewal in transform space, and AW/Cauchy-FFT inversion.
- **Time-domain recursion family**: exact propagation on the transient state space, including sparse absorbing recursion and absorbed pair-distribution updates.

The main benchmark is not a forced full-tail comparison. It is a report-production comparison: how much time does each family need to deliver the quantity that the scientific report actually required?

2 Unified Notation and Runtime Protocol

Let X_t be a discrete-time Markov chain, T a target set, $\tau_T = \inf\{t \geq 1 : X_t \in T\}$ the first-passage time,

$$f_T(t) = \mathbb{P}(\tau_T = t), \tag{1}$$

$$S_T(t) = \mathbb{P}(\tau_T > t), \tag{2}$$

$$\widetilde{f}_T(z) = \sum_{t \geq 1} f_T(t) z^t, \tag{3}$$

$$\widetilde{P}_{ij}(z) = \sum_{t \geq 0} \mathbb{P}_i(X_t = j) z^t = \left[(I - zP)^{-1}\right]_{ij}. \tag{4}$$

For single-target problems, renewal closes immediately:

$$\widetilde{f}_{i\to m}(z) = \frac{\widetilde{P}_{im}(z)}{\widetilde{P}_{mm}(z)}. \tag{5}$$

For multiple targets, renewal becomes a small linear system on target-to-target propagators. All timings use single-thread BLAS, one warm-up run, three measured runs, and the median wall time as the reported benchmark value.

3 What Luca/GF Means Here

Throughout this report, 'Luca/GF' is a family label rather than the name of one single solver. The family bundles four layers that appear in different proportions across the six workloads [1, 2, 3, 4]:

1. a closed-form or defect-free propagator;
2. a finite-dimensional defect or heterogeneity correction on a selected support;
3. a one-target or multi-target renewal closure;
4. AW/Cauchy-FFT coefficient recovery of the FPT generating function [5].

This distinction matters because not every workload uses all four layers equally. In particular, the two-target C1 and LF1 benchmarks use selected-propagator recovery rather than a dense global inverse over every transient state, while ENC-FIXED is a $\beta = 0$ control that tests the pair-propagator layer without a shortcut-defect correction.

4 Luca / Generating-Function Family: Mathematical Spine

The common transform-domain workflow is:

1. Build a defect-free propagator or a closed-form single-walker PGF.
2. Recover the defected propagator through a finite-dimensional Woodbury or determinant system.
3. Close a one-target or multi-target renewal relation to obtain the FPT generating function.
4. Invert the PGF numerically with an AW/Cauchy-FFT contour.

The inversion step evaluates

$$f(t) = [z^t]\widetilde{f}(z) = \frac{1}{2\pi i}\oint_{|z|=r} \frac{\widetilde{f}(z)}{z^{t+1}}\,dz \approx \frac{r^{-t}}{m}\sum_{k=0}^{m-1} \widetilde{f}\Big(re^{2\pi ik/m}\Big)\, e^{-2\pi ikt/m}. \tag{6}$$

This is the shared numerical shell behind the ring, encounter, two-target, and reflecting GF workloads [5].

4.1 Encounter-Specific GF Route

For the two-walker shortcut encounter, the defect-free pair propagator is

$$G_0\big((n_0, m_0) \to (x, y); z\big) = \frac{1}{N^2}\sum_{k_1,k_2} \frac{h_{k_1}(x, n_0)h_{k_2}(y, m_0)}{1 - z\lambda_{k_1}^{(1)}\lambda_{k_2}^{(2)}}. \tag{7}$$

The shortcut becomes a line defect on the pair torus and is restored by

$$G(z) = G_0(z) + z\, G_0(z) U \left(I - z V^\top G_0(z) U \right)^{-1} V^\top G_0(z). \tag{8}$$

A1 and A8 are therefore not the production solver for the encounter report; they are the spectral anchor from which the new timed GF route is constructed.

5 Time-Domain Recursion Family: Mathematical Spine

The time-domain family works directly with absorbing transient updates. If Q is the transient transition matrix and r the transient-to-target flux vector, then

$$u_{t+1} = u_t Q, \tag{9}$$
$$f(t+1) = u_t r, \tag{10}$$
$$S(t+1) = u_{t+1} \mathbf{1}. \tag{11}$$

For two-target problems, r becomes a two-column target flux block. For the pair encounter, the exact recursion is naturally written in matrix form:

$$J_{t+1} = P_1^\top J_t P_2, \tag{12}$$
$$f_{\text{enc}}(t+1) = \operatorname{tr}(J_{t+1}), \tag{13}$$

with the diagonal or designated encounter site zeroed after each step to enforce absorption. No asymptotic approximation is used here; the family name refers only to the domain in which the update is carried out.

6 Workload Instantiations and Configuration Figures

Each workload is documented here with a detailed configuration figure, a workload-specific specialization of the two solver families, and a provenance-aware description of what the transform-domain route really means in that case.

workload	model	states	defects	targets
RING-1T-paper	ring_single_target	100	1	1
ENC-FIXED	ring_encounter_fixed	900	0	1
ENC-ANY	ring_encounter_any	10201	101	101
TT-C1	grid2d_two_target	1679	632	2
TT-LF1	grid2d_two_target	1679	2	2
REF-S0	grid2d_reflecting	840	9	1

Table 1: The six benchmark workloads kept in the unified comparison.

6.1 RING-1T-paper

This workload is the analytic single-target ring benchmark. In this case the Giuggioli-style transform-domain family really is the clean closed-form route: a defect-free ring propagator, a

rank-one shortcut-column correction, and scalar renewal to the target [1, 5]. The specialized formulas are

$$\widetilde{P}_{n_0\to n}(u) = \sum_{k=0}^{N-1} \frac{h_k(n, n_0)}{1 - u\alpha_k}, \tag{14}$$

$$\widetilde{F}_{n_0\to m}(u) = \frac{\widetilde{P}_{n_0\to m}(u)}{\widetilde{P}_{m\to m}(u)}, \tag{15}$$

followed by AW/Cauchy coefficient recovery [1, 5]. The time solver instead iterates the absorbing chain directly and serves as the exact baseline for the same observable. Direct speed mapping for the configuration figure: the diagnostic task is faster with Luca/GF (time=0.007141s, luca=0.0006551s, 10.9x separation). The curve task keeps the same winner (time=0.01741s, luca=0.002149s).

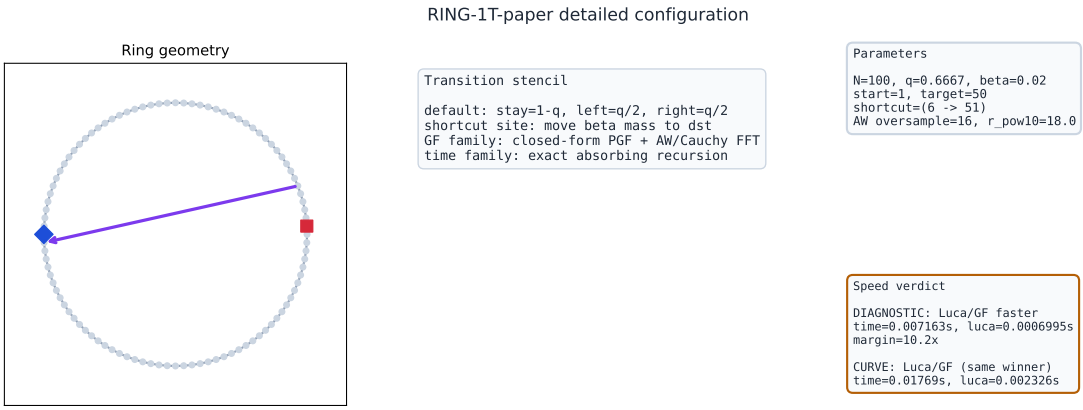

Figure 1: Detailed ring geometry for the paper-like single-target shortcut benchmark.

6.2 ENC-FIXED

The fixed-site encounter workload is a single-target first-passage problem on the pair chain. The benchmark configuration here is a control with $\beta = 0$, so this is *not* a shortcut-defect production case. The correct family-level description is: defect-free pair propagator plus one-target renewal on the pair torus [1, 5]. The benchmark formulas are therefore

$$G_0\big((n_0, m_0) \to (x, y); z\big) = \frac{1}{N^2} \sum_{k_1,k_2} \frac{h_{k_1}(x, n_0) h_{k_2}(y, m_0)}{1 - z\lambda_{k_1}^{(1)} \lambda_{k_2}^{(2)}}, \tag{16}$$

$$\widetilde{f}_\delta(z) = \frac{G\big((n_0, m_0) \to (\delta, \delta); z\big)}{G\big((\delta, \delta) \to (\delta, \delta); z\big)}. \tag{17}$$

The time family evolves the absorbed pair density exactly on the N^2 state space. This workload tests the pair-propagator layer of the family, not the finite-defect shortcut correction layer. Direct speed mapping for the configuration figure: the diagnostic task is faster with Time recursion (time=0.006613s, luca=0.3216s, 48.6x separation). The curve task keeps the same winner (time=0.006537s, luca=0.3167s).

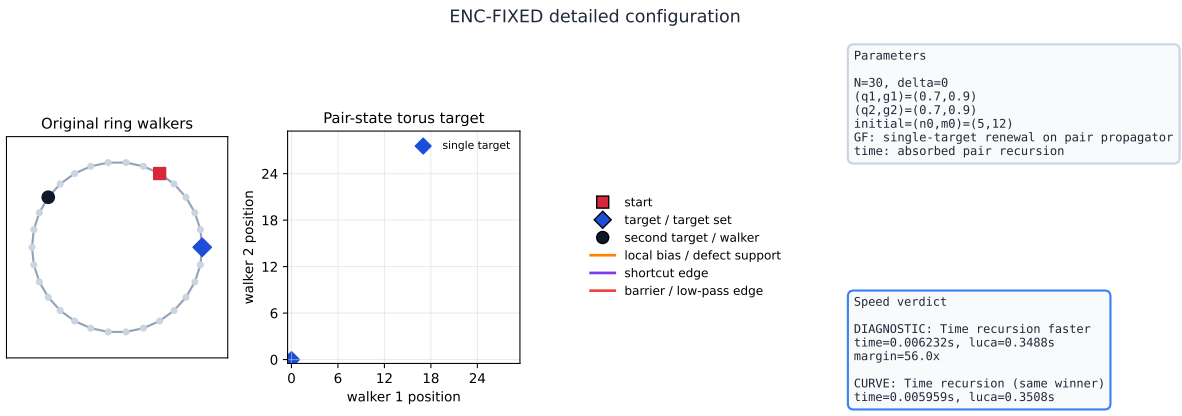

Figure 2: Detailed configuration for the fixed-site encounter benchmark, including the single pair-state target.

6.3 ENC-ANY

The anywhere-encounter workload promotes the target from one pair-state to the full diagonal set $D = \{(x, x) : x \in \mathbb{Z}_N\}$. Here the shortcut becomes a line defect on the pair torus, so the transform-domain route is: defect-free pair propagator, finite-dimensional defect restoration, and multi-target renewal on the diagonal set [1, 4, 5]. The benchmark formulas are

$$G(z) = G_0(z) + z\, G_0(z) U \left(I - z V^\top G_0(z) U \right)^{-1} V^\top G_0(z), \tag{18}$$

$$\widetilde{\mathbf{f}}_D(z) = G_{DD}(z)^{-1} G_{Ds}(z), \tag{19}$$

$$\widetilde{f}_{\mathrm{enc}}(z) = \mathbf{1}^\top \widetilde{\mathbf{f}}_D(z). \tag{20}$$

This is the proper place in the report to talk about a shortcut-induced defect line and a finite-dimensional resolvent correction. Direct speed mapping for the configuration figure: the diagnostic task is faster with Time recursion (time=0.04208s, luca=3.282s, 78.0x separation). The curve task keeps the same winner (time=0.07917s, luca=6.586s).

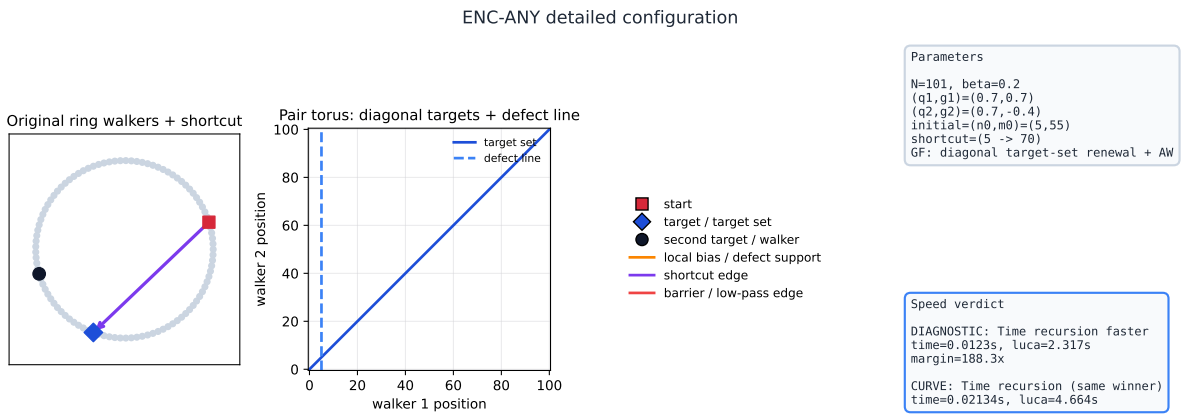

Figure 3: Detailed configuration for the anywhere-encounter benchmark, including the diagonal target set and the shortcut-induced defect line.

6.4 TT-C1

C1 is the practical two-target corridor benchmark. The transform-domain solver used in the benchmark is intentionally narrower than a full dense resolvent solve: it recovers only the selected propagators needed by the two-target renewal closure, with the heterogeneity compressed to the local support [2, 3, 4, 5]. To avoid overstating the method, the benchmark should therefore be described by

$$G_{sT}(z) = G_{sT}^{(0)}(z) + \text{defect-reduced correction on the selected support}, \tag{21}$$

$$\widetilde{\mathbf{f}}(z) = G_{TT}(z)^{-1} G_{Ts}(z), \qquad T = \{m_1, m_2\}, \tag{22}$$

rather than by a claim that the benchmark forms the full matrix $(I - zQ)^{-1}$ over every transient state. The time solver is the sparse exact absorbing recursion on the transient graph, continued to the long horizon actually needed by the scientific claim. Direct speed mapping for the configuration figure: the diagnostic task is faster with Time recursion (time=0.01337s, luca=74.54s, 5575.7x separation). The curve task keeps the same winner (time=0.06406s, luca=74.17s).

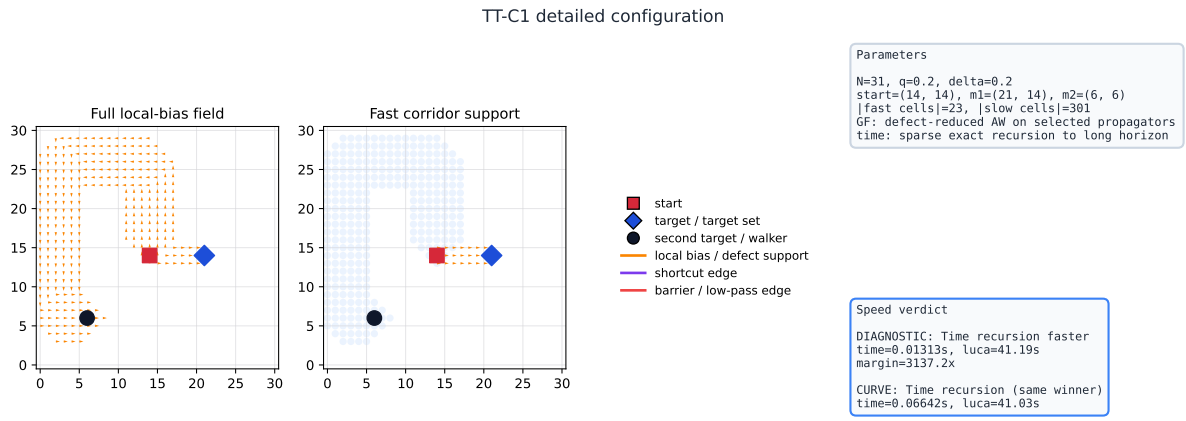

Figure 4: Detailed C1 geometry, showing both the full local-bias field and the corridor support that determines the defect-reduced dimension.

6.5 TT-LF1

TT-LF1 keeps the same two-target selected-propagator mathematics as C1, but collapses the heterogeneity support to an ultra-sparse perturbation [2, 3, 4, 5]. It is therefore the in-repo positive anchor where the defect-reduced transform-domain route can beat long-horizon sparse recursion, not because it uses a different theory, but because the same theory is operating in a much smaller support regime. Direct speed mapping for the configuration figure: the diagnostic task is faster with Luca/GF (time=0.1881s, luca=0.01722s, 10.9x separation). The curve task keeps the same winner (time=0.1882s, luca=0.01771s).

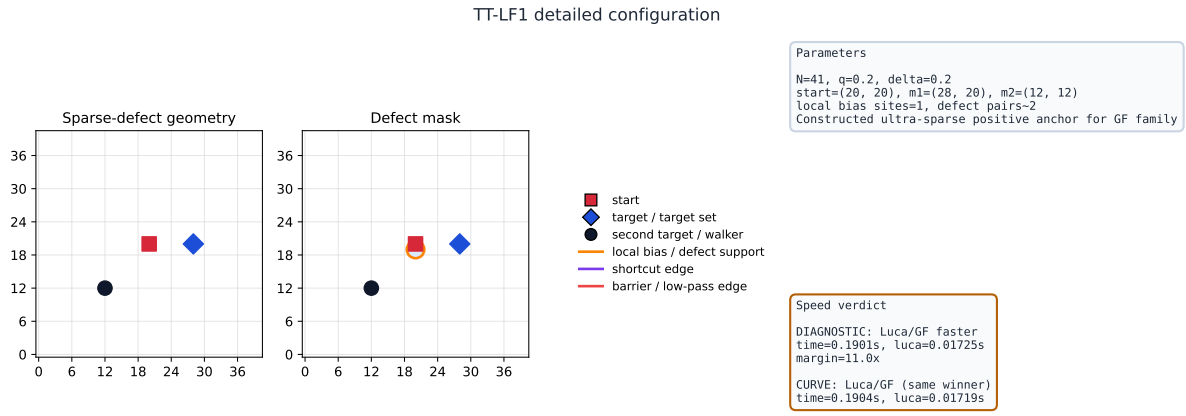

Figure 5: Detailed LF1 sparse-defect geometry used as the positive Luca/GF anchor.

6.6 REF-S0

REF-S0 is the reflecting low-defect control. The transform-domain route remains a single-target reflecting-lattice renewal, and the full AW inversion is still numerically feasible only because the defect count is deliberately kept small [2, 3, 4, 5]:

$$\widetilde{F}_{s\to m}(z) = \frac{G_{sm}(z)}{G_{mm}(z)}, \tag{23}$$

$$f(t) = [z^t]\widetilde{F}_{s\to m}(z). \tag{24}$$

The time family is the standard exact absorbing recursion on the reflecting lattice. The role of this workload is to mark the low-defect edge of feasibility; it should not be described as evidence that full AW remains the default method for generic 2D heterogeneous cases. Direct speed mapping for the configuration figure: the diagnostic task is faster with Time recursion (time=0.005605s, luca=0.0724s, 12.9x separation). The curve task keeps the same winner (time=0.01666s, luca=0.2829s).

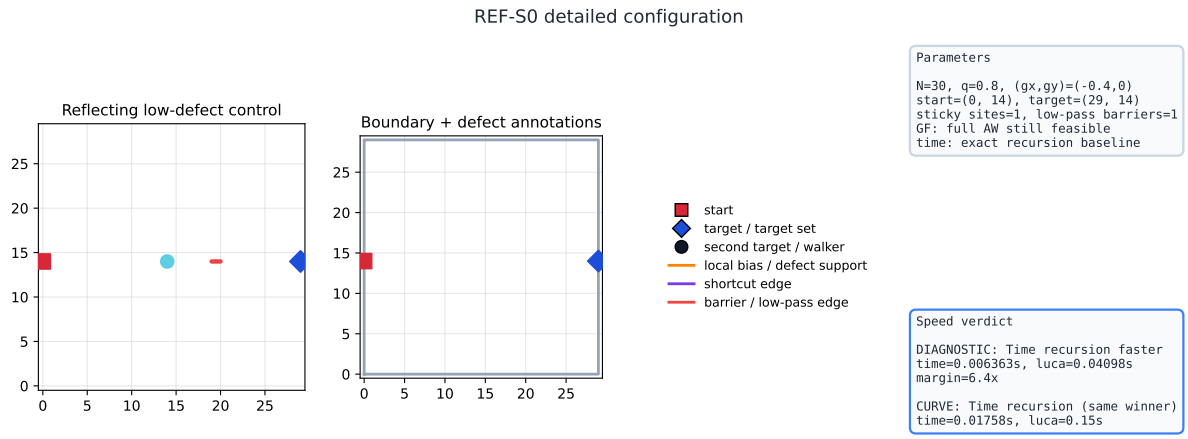

Figure 6: Detailed reflecting low-defect S0 control geometry, highlighting the sticky site and the partially permeable barrier.

7 Main Benchmark Results

workload family	Luca/GF family	time recursion family	comment
RING-1T-paper	$O(mN) + O(m \log m)$	$O(TN)$	closed-form single-target PGF plus AW
ENC-FIXED	$O(mN^3)$	$O(TN^3)$	single-target renewal on the pair chain
ENC-ANY	$O(mN^3)$	$O(TN^3)$	diagonal target set with $\lvert D\rvert = N$
TT-C1 / TT-LF1	$O(mM^3) + O(m \log m)$	$O(T\lvert E\rvert)$	defect-reduced AW vs sparse exact recursion
REF-S0	$O(mn_T^3) + O(m \log m)$	$O(T\lvert E\rvert)$	low-defect full AW vs exact recursion

Table 2: Family-level complexity summary.

workload	source report	Luca/GF solver	time solver
RING-1T-paper	ring_lazy_flux	ring_analytic_aw	ring_time_absorption
ENC-FIXED	ring_two_walker_encounter_shortcut	encounter_fixedsite_gf_aw	pair_fixedsite_time_recursion
ENC-ANY	ring_two_walker_encounter_shortcut	encounter_anywhere_gf_aw	pair_time_recursion
TT-C1	grid2d_two_target_double_peak	two_target_defect_reduced_aw	two_target_sparse_exact
TT-LF1	grid2d_two_target_double_peak	two_target_defect_reduced_aw	two_target_sparse_exact
REF-S0	grid2d_reflecting_bimodality	reflecting_full_aw	reflecting_exact_recursion

Table 3: Family-internal solver mapping for the six workloads.

One-glance overview: the numbered tiles 1-6 match the numbered workload groups on the runtime chart

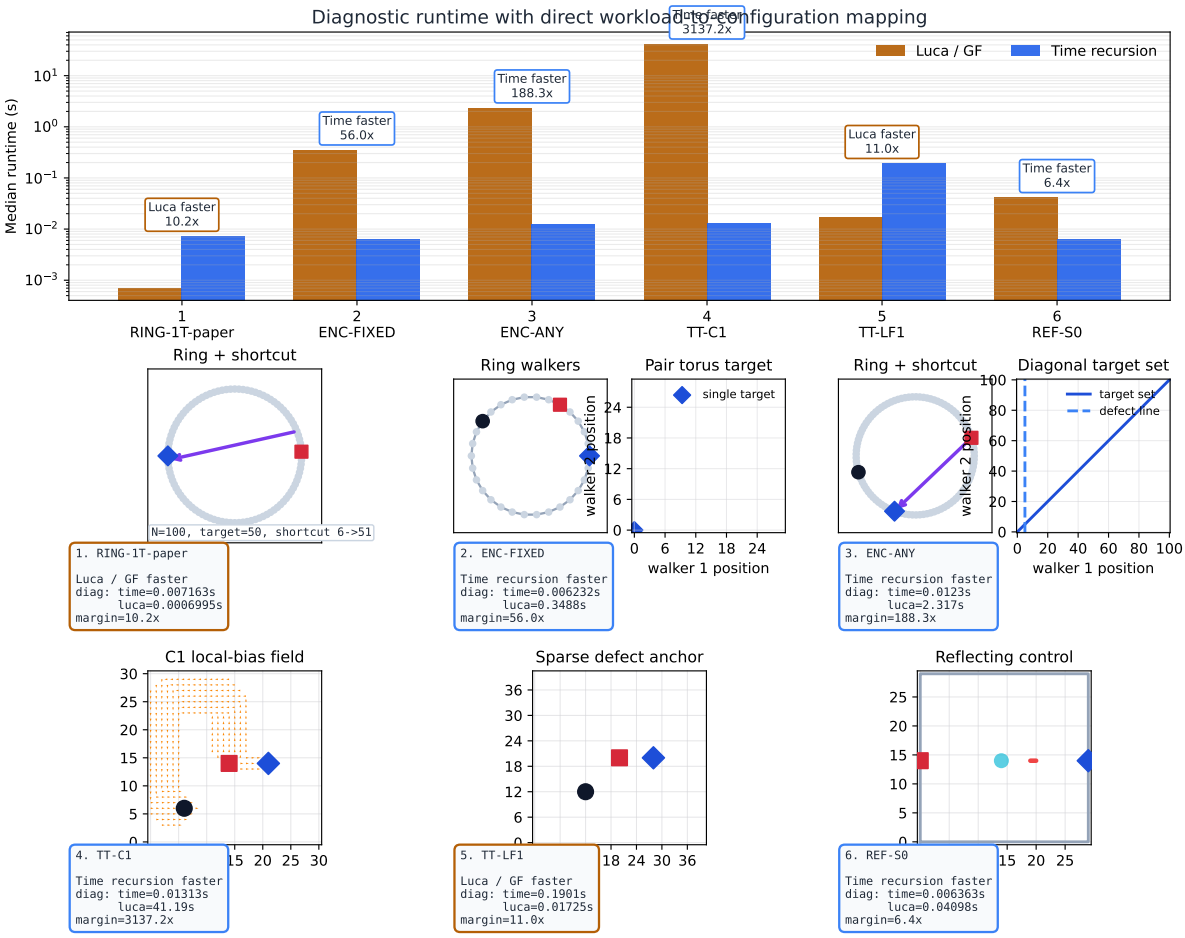

Figure 7: One-glance benchmark overview. The numbered mini-panels 1–6 are the six workload configurations, and the numbered groups on the top runtime chart follow the same order. This is the main practical benchmark figure used in the report.

workload	Luca/GF (s)	time recursion (s)	effective T (L/T)	recommended	L^1
RING-1T-paper	0.0006551	0.007141	800/800	Luca / generating-function family	1.643e-06
ENC-FIXED	0.3216	0.006613	900/900	Time-domain recursion family	1.382e-10
ENC-ANY	3.282	0.04208	420/420	Time-domain recursion family	9.020e-14
TT-C1	74.54	0.01337	800/1200	Time-domain recursion family	9.912e-10
TT-LF1	0.01722	0.1881	80/12000	Luca / generating-function family	2.135e-10
REF-S0	0.0724	0.005605	300/300	Time-domain recursion family	1.671e-09

Table 4: Diagnostic-task runtime table.

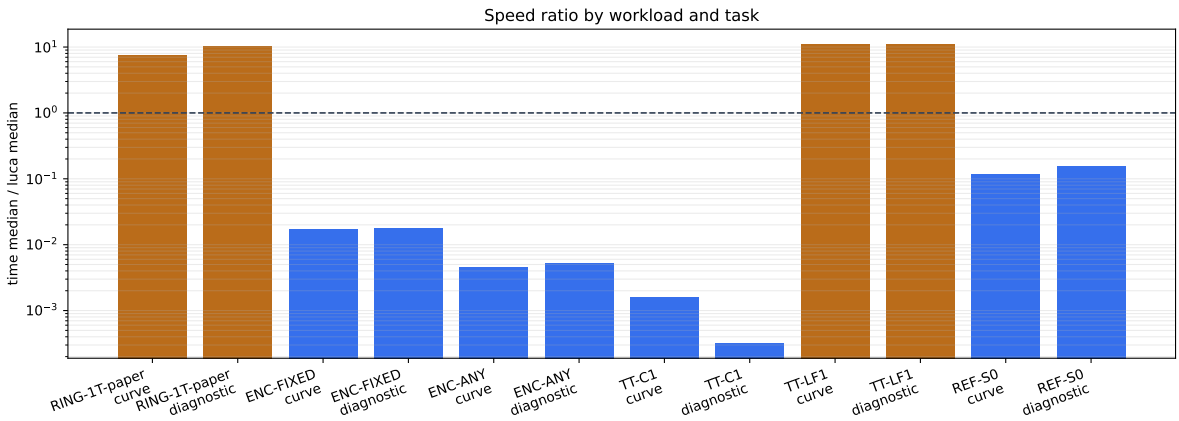

Figure 8: Per-workload ratio $t_{\text{time}}/t_{\text{Luca}}$. Ratios above one favor the Luca/GF family.

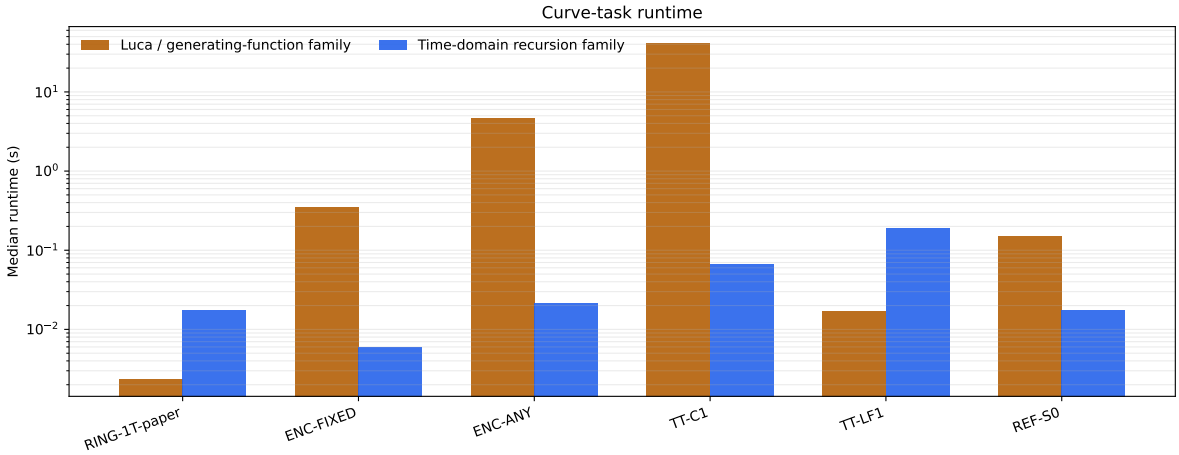

Figure 9: Curve-task runtime, kept secondary so that long-tail costs do not overwrite the practical report-production benchmark.

workload	Luca/GF (s)	time recursion (s)	effective T (L/T)	recommended	L^1
RING-1T-paper	0.002149	0.01741	2000/2000	Luca / generating-function family	7.588e-10
ENC-FIXED	0.3167	0.006537	900/900	Time-domain recursion family	1.382e-10
ENC-ANY	6.586	0.07917	800/800	Time-domain recursion family	1.045e-13
TT-C1	74.17	0.06406	800/6000	Time-domain recursion family	9.912e-10
TT-LF1	0.01771	0.1882	80/12000	Luca / generating-function family	2.135e-10
REF-S0	0.2829	0.01666	1200/1200	Time-domain recursion family	3.005e-09

Table 5: Curve-task runtime table.

8 Method Boundaries

The benchmark supports a narrower claim than a generic 'Luca/GF versus exact recursion' slogan. What it actually resolves is where the transform-domain family remains a practical production route once the right mathematical layer is identified. The boundary lines are:

- use the transform-domain family aggressively when a clean closed form already exists (RING-1T-paper),
- keep it as a valid but usually slower validation route when the pair-propagator formalism is correct but the defect support is no longer tiny (ENC-ANY),

- describe it carefully as selected-propagator recovery, not full dense resolvent inversion, in bounded heterogeneous two-target problems (TT-C1 and TT-LF1),
- treat low-defect full AW controls as feasibility markers rather than default production solvers for generic 2D heterogeneous cases (REF-S0).

9 Recommendation Matrix

Scenario	Recommendation
Single-target ring with analytic PGF	Prefer Luca/GF when the closed form already exists and longer windows are needed.
Shortcut encounter (anywhere/fixed-site)	Prefer time recursion for production figures; use GF when transform-domain renewal validation is itself the task.
2D two-target, medium/high defect	Prefer time recursion; C1-like cases still punish the defect-reduced AW route.
2D two-target, ultra-sparse defect	Luca/GF can win; TT-LF1 is the in-repo positive anchor.
2D reflecting low-defect control	Time recursion still wins on wall time in this repo.

Table 6: Decision matrix derived from the six-workload benchmark.

10 Conclusion

The comparison line is now fully centralized in one report. The scientific reports remain focused on science, while this document owns the algorithmic comparison, the fairness rule, the workload inventory, and the implementation-level guidance for choosing between transform-domain and time-domain solvers.

A Appendix A: Symbol Table and Variable Mapping

$$P : \text{full transition operator}, \qquad Q : \text{transient transition operator}, \tag{25}$$

$$r : \text{transient-to-target flux}, \qquad U\Delta V^\top : \text{defect update}, \tag{26}$$

$$G_0(z) : (I - zQ_0)^{-1}, \qquad G(z) : (I - zQ)^{-1}, \tag{27}$$

$$m : \text{AW FFT length}, \qquad r : \text{AW contour radius}. \tag{28}$$

B Appendix B: Detailed GF Derivations

The GF family reduces every workload to the same algebraic template. Starting from a defect-free Q_0, write

$$Q = Q_0 + U\Delta V^\top, \tag{29}$$

$$I - zQ = (I - zQ_0)\Big[I - z(I - zQ_0)^{-1}U\Delta V^\top\Big], \tag{30}$$

$$(I - zQ)^{-1} = (I - zQ_0)^{-1} + z(I - zQ_0)^{-1}U\Big[I - zV^\top(I - zQ_0)^{-1}U\Delta\Big]^{-1}V^\top(I - zQ_0)^{-1}. \tag{31}$$

For a target set $T = \{m_1, \ldots, m_K\}$, renewal is written as

$$G_{sT}(z) = \widetilde{\mathbf{f}}_T(z)^\top G_{TT}(z), \tag{32}$$

$$\widetilde{\mathbf{f}}_T(z) = G_{TT}(z)^{-1}G_{Ts}(z). \tag{33}$$

The one-target formula is the $K = 1$ specialization of this matrix equation.

C Appendix C: Detailed Time-Recursion Derivations

Given the transient state vector u_t, exact recursion uses

$$u_0 = e_s, \tag{34}$$

$$u_{t+1} = u_t Q, \tag{35}$$

$$f(t+1) = u_t r, \tag{36}$$

$$S(t+1) = u_{t+1}\mathbf{1}. \tag{37}$$

For two-target problems, let $R = [r^{(1)}\ r^{(2)}]$ and then $f_1(t+1) = u_t r^{(1)}$, $f_2(t+1) = u_t r^{(2)}$. For pair encounters, the matrix recursion form is numerically convenient because the encounter condition is a structured absorbing set on the pair torus.

D Appendix D: Per-Workload Formula Specialization and Audit Map

RING-1T-paper uses a scalar renewal. ENC-FIXED uses a single-target renewal on the pair chain. ENC-ANY uses a diagonal target-set renewal on the pair torus. TT-C1 and TT-LF1 use a two-target renewal after defect-reduced selected-propagator recovery. REF-S0 uses the single-target reflecting resolvent. These are exactly the formulas instantiated in Section 5 alongside the detailed configuration figures.

workload	solver pair	mathematical object	paper source	implementation anchor
RING-1T-paper	ring_analytic_aw vs ring_time_absorption	Single-target renewal from a closed-form defect-free ring propagator with a rank-one shortcut-column correction.	Sarvaharman & Giuggioli, Phys. Rev. E 102, 062124 (2020); Abate & Whitt, INFORMS J. Comput. 18(4), 408-421 (2006)	‘vkcore.ring.jumpover_pipeline.aw_first_ab implements the transform-domain route; ‘exact_first_absorption_pmf‘ is the absorbing-chain baseline.
ENC-FIXED	encounter_fixedsite_gf_aw vs pair_fixedsite_time_recursion	Single-target renewal on the pair torus, specialized to the fixed encounter site $(ndelta, ndelta)$.	Sarvaharman & Giuggioli, Phys. Rev. E 102, 062124 (2020); Abate & Whitt, INFORMS J. Comput. 18(4), 408-421 (2006)	‘vkcore.ring.encounter.encounter_gf_fixed evaluates the fixed-site transform-domain solver; ‘encounter_time_fixed_site‘ is the exact absorbed pair recursion.
ENC-ANY	encounter_anywhere_gf_aw vs pair_time_recursion	Diagonal-target renewal on the pair torus with a shortcut-induced defect line.	Sarvaharman & Giuggioli, Phys. Rev. E 102, 062124 (2020); Giuggioli, Sarvaharman, Das, Marris & Kay, arXiv:2311.00464 (2023); Abate & Whitt, INFORMS J. Comput. 18(4), 408-421 (2006)	‘vkcore.ring.encounter.encounter_gf_anyw builds the pair-kernel, restores the defected selected propagators, and closes diagonal-target renewal; ‘encounter_time_anywhere‘ is the exact absorbed pair recursion.
TT-C1	two_target_defect_reduced_aw vs two_target_sparse_exact	Two-target renewal from defect-reduced selected propagators in a bounded heterogeneous lattice, not a full dense resolvent of all transient states.	Das, Giuggioli et al., Phys. Rev. Research 5, 043281 (2023); Das & Giuggioli, J. Stat. Mech. 013201 (2023); Giuggioli, Sarvaharman, Das, Marris & Kay, arXiv:2311.00464 (2023); Abate & Whitt, INFORMS J. Comput. 18(4), 408-421 (2006)	‘vkcore.comparison.run_two_target_defect performs selected-propagator recovery plus AW inversion; ‘run_exact_two_target‘ is the sparse absorbing recursion on the transient graph.
TT-LF1	two_target_defect_reduced_aw vs two_target_sparse_exact	The same two-target selected-propagator renewal as C1, but in an ultra-sparse heterogeneity regime where the reduced support is genuinely tiny.	Das, Giuggioli et al., Phys. Rev. Research 5, 043281 (2023); Das & Giuggioli, J. Stat. Mech. 013201 (2023); Giuggioli, Sarvaharman, Das, Marris & Kay, arXiv:2311.00464 (2023); Abate & Whitt, INFORMS J. Comput. 18(4), 408-421 (2006)	The same code path as C1 (‘run_two_target_defect_reduced_aw‘ versus ‘run_exact_two_target‘), but with the LF1 sparse support geometry from ‘build_luca_fast_geometry‘.
REF-S0	reflecting_full_aw vs reflecting_exact_recursion	Single-target reflecting-lattice renewal in a low-defect control where the full AW route is still numerically feasible.	Das, Giuggioli et al., Phys. Rev. Research 5, 043281 (2023); Das & Giuggioli, J. Stat. Mech. 013201 (2023); Giuggioli, Sarvaharman, Das, Marris & Kay, arXiv:2311.00464 (2023); Abate & Whitt, INFORMS J. Comput. 18(4), 408-421 (2006)	‘vkcore.comparison.run_reflecting_full_aw is the low-defect transform-domain control, while ‘run_reflecting_exact_recursion‘ is the exact absorbing recursion baseline.

Table 7: Audit appendix: actual solver pair, mathematical object, paper provenance, and implementation anchor for each workload.

E Appendix E: Complexity, Memory, Error Metrics, and Benchmark Protocol

The runtime comparison is accepted only after a same-window numerical agreement check between the two families. For each workload, the overlap window is reported through L^1, L^∞, and a peak-location consistency flag. This means wall time is never reported without a paired numerical sanity check.

F Appendix F: Embedded Historical Full-FPT Note

The repository no longer keeps an external active compare report for the old fixed full-tail fairness question. Instead, the historical note is embedded here so that the active comparison line remains single-source.

quantity	value
diagnostic median t_{time}/t_{Luca}	0.04899
curve median t_{time}/t_{Luca}	0.03976
historical policy retained here	embedded appendix only
external compare report	retired

Table 8: Embedded historical fairness note retained inside the unified report.

References

References

[1] Sarvaharman & Giuggioli, Phys. Rev. E 102, 062124 (2020). Available at https://research-information.bris.ac.uk/ws/files/267741195/PhysRevE.102.062124.pdf.

[2] Das, Giuggioli et al., Phys. Rev. Research 5, 043281 (2023). Available at https://journals.aps.org/prresearch/abstract/10.1103/PhysRevResearch.5.043281.

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[4] Giuggioli, Sarvaharman, Das, Marris & Kay, arXiv:2311.00464 (2023). Available at https://research-information.bris.ac.uk/en/publications/multi-target-search-in-bounded-and-heterogeneous-environments-a-l/.

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